

Full-Cohort Revised Thesis

MIMIC-IV v3.1 treatment-feasibility modelling after inclusion/exclusion criteria

Abstract

This revised thesis summary reports a full-cohort MIMIC-IV v3.1 modelling study after applying the study inclusion and exclusion criteria. The final modelling dataset contains 32,399 adult first ICU stays, not the previous 2,000-stay benchmark subset. The strongest rerun model was Stacked Classical with AUROC 0.755.

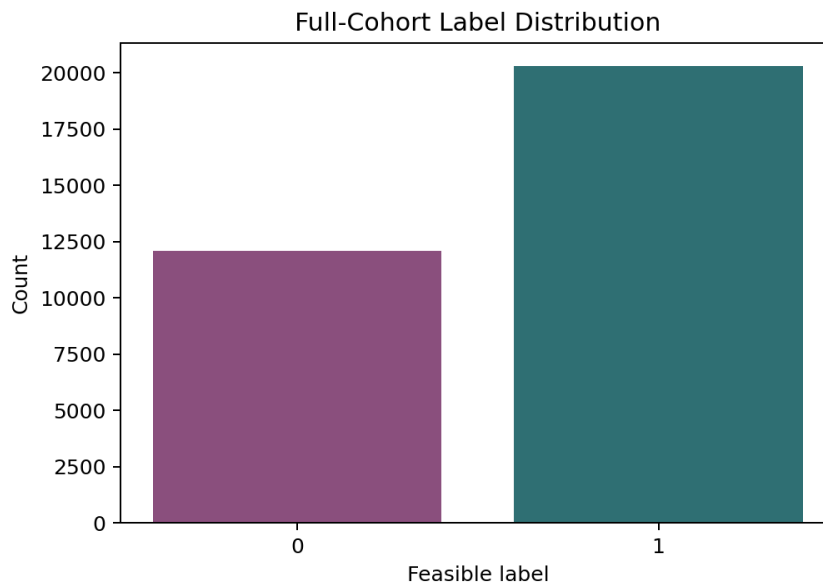
Methods

The cohort included adult first ICU stays with ICU length of stay at least 4 hours, non-missing discharge location, and an available primary ICD-10 diagnosis. First-24-hour laboratory and vital-sign features were extracted from the full eligible cohort. Missing feature values were quantified before median imputation and tokenization.

Quantity	Value
Eligible adult first ICU stays	32,399
Final modelling rows	32,399
Unique ICU stays	32,399
Unique subjects	32,399
Token tensor shape	(32399, 25, 4)
All eligible data used	yes

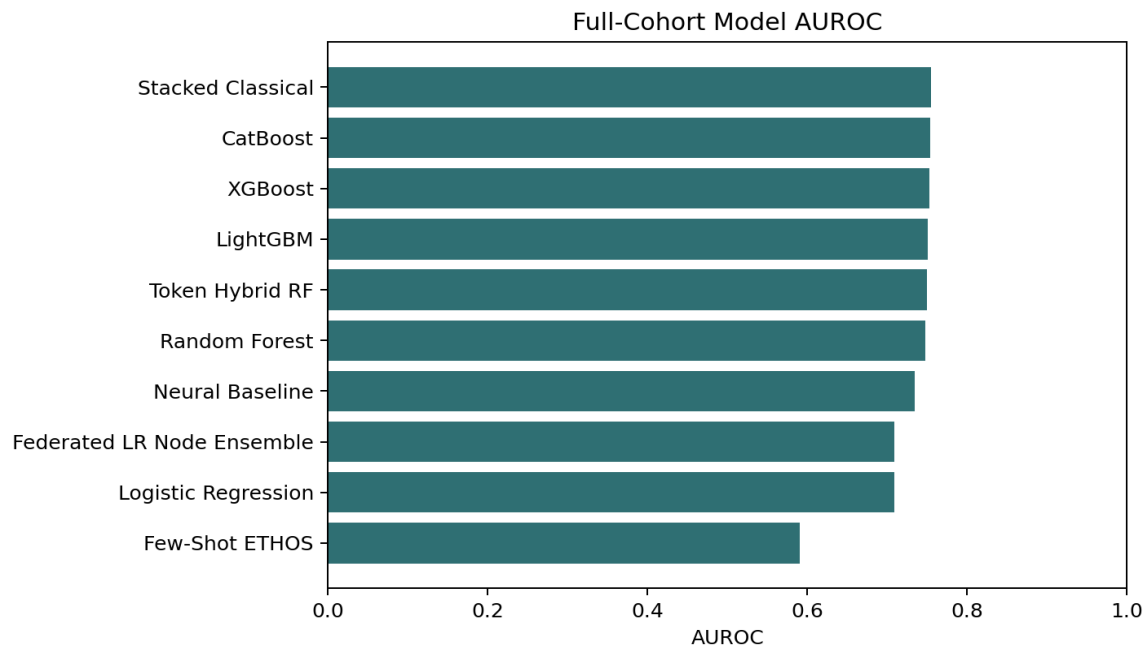
Results

Label distribution was 0: 12102, 1: 20297. Mean pre-imputation feature missingness was 25.2%; post-imputation missing cells in the modelling matrix were 0.



Model	AUROC	95% CI	F1	AUPRC	Accuracy	MCC
Stacked Classical	0.755	0.743-0.766	0.677	0.826	0.697	0.354
CatBoost	0.753	0.741-0.764	0.673	0.824	0.700	0.348
XGBoost	0.753	0.740-0.764	0.675	0.825	0.695	0.350
LightGBM	0.751	0.739-0.762	0.674	0.823	0.690	0.348
Token Hybrid RF	0.750	0.738-0.762	0.671	0.821	0.695	0.343
Random Forest	0.748	0.735-0.760	0.667	0.819	0.685	0.333

Model	AUROC	95% CI	F1	AUPRC	Accuracy	MCC
Neural Baseline	0.735	0.722-0.747	0.651	0.809	0.656	0.324
Federated LR Node Ensemble	0.709	0.696-0.722	0.656	0.778	0.678	0.311
Logistic Regression	0.709	0.696-0.722	0.656	0.778	0.680	0.312
Few-Shot ETHOS	0.590	0.576-0.604	0.569	0.668	0.588	0.140



Discussion

The revised framing is a full-cohort modelling study. The old 2,000-stay benchmark is not used as evidence for the main full-cohort claims and should only be cited as historical benchmark context.

Limitations

The feasibility outcome remains a proxy derived from discharge disposition, survival flag, and ICU length of stay. Feature extraction uses selected first-24-hour labs and vitals, and unmeasured features are imputed after missingness is recorded. Model results are single-split reruns unless otherwise stated.